

Freight Rate Prediction – Validation & Split Report

1. Data split and validation approach

Key observation: train_test.csv covers 2025-01-01 to 2025-10-31 (48,000 loads) while validation.csv covers 2025-11-01 to 2025-12-31 (12,000 loads). The final task is therefore a *forecast into the future*, not an interpolation. A random train/test split would leak same-day and same-lane information and overstate accuracy, so I used **expanding-window, forward-chaining validation**: each fold trains only on data before the fold and tests on the following period (Jul–Aug, Sep, Oct). Hyper-parameters and feature choices were selected on these folds; the final model is refit on all 48,000 labeled loads.

Fold results (model vs. naive baseline = equipment-median \$/mile × distance):

Fold	Train n	Test n	MAE \$	MAPE %	MedAPE %	Naive MAE \$	Naive MAPE %
2025-07-01..2025-08-31	28806	9671	103.9	4.41	1.90	222.7	10.11
2025-09-01..2025-09-30	38477	4670	96.2	3.96	1.59	227.3	10.04
2025-10-01..2025-10-31	43147	4853	99.9	4.68	1.49	231.3	10.93
Mean			100.0	4.35	1.66	227.1	10.36

RMSE (~\$628) is dominated by ~1.5% of loads with extreme rates (up to ~\$25k; rate-per-mile residuals beyond ±65%) that are not predictable from the features; hence an L1 objective and MAE/MAPE as the primary metrics.

2. Data-quality findings and handling

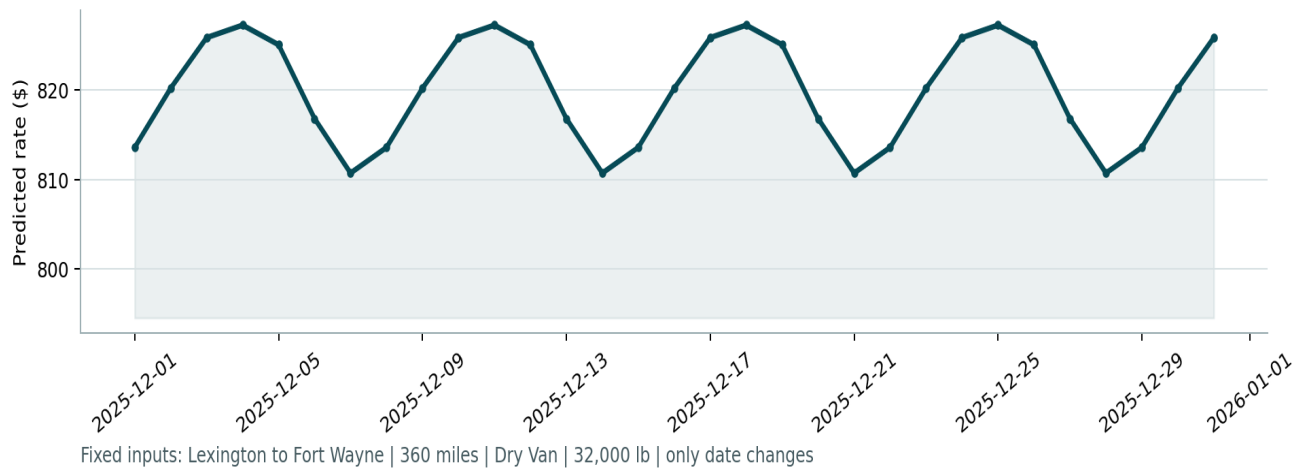
- **Weight:** 292 negative values (magnitudes plausible) treated as sign errors → absolute value; 300 missing → kept as NaN with a missing-indicator.
- **market_index:** 374 missing; it is a daily level with ±0.025 per-row noise.
- **Distance:** per-route distances vary (std ≈ 19 mi, consistent with road-distance noise) and are floored at 70 miles, which makes the road/great-circle ratio look extreme on very short lanes; these are valid and were kept.
- **Label outliers:** ~1.5% of rates deviate >65% from comparable loads; handled with an L1 loss rather than deleting rows.
- No duplicate loads; 64 locations, no spelling variants.

3. Model choice

Rate scales roughly linearly with distance, so I model $\log(\text{\$/mile})$ and multiply back by distance. LightGBM captures distance curvature (short-haul premium), equipment and lane geography (lat/lon) with little tuning; results were insensitive to hyper-parameters (MAE \$99.9–100.5). **market_index was removed**: it improved in-sample fit but, out of time, worsened MAE from ~\$100 to ~\$145 with a -2% to -6% bias, because its relationship to rate is not stable across seasons. **quote_signal** was removed since its correlation with the residual flips sign month to month (+ in Jan–Mar, Jun, Sep; – in Apr–May, Jul, Oct).

4. Fixed December prediction chart (produced by score.py)

Candidate: December 2025 Predicted Load Rate



Caveat: the December inputs have no market or quote values and the model has no seasonal-trend feature, so only the day-of-week pattern (~\$811–\$827, $\pm 1\%$) is expressed. This is deliberate: date-trend features cannot extrapolate beyond the training months and were unstable out of time. The level therefore reflects the Jan–Oct average for this lane; any genuine December seasonality is not modeled.